

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Ionic and Covalent Compound I IC1-LO1

Atoms are especially stable when their valence shell contains _____ electrons.

- A. 2
- B. 6
- C. 8
- D. 18

A2 Topic: Ionic and Covalent Compound I IC1-LO2

A calcium atom (Group 2A) forms the ion _____.

- A. Ca^+
- B. Ca^{2+}
- C. Ca^{2-}
- D. Ca^-

A3 Topic: Ionic and Covalent Compound I IC1-LO2

Which pair of species is isoelectronic?

- A. Na^+ and Ne
- B. Na^+ and Ar
- C. Cl^- and Ne
- D. K^+ and Ne

A4 Topic: Ionic and Covalent Compound I IC1-LO3

The correct formula for the compound of Al^{3+} and O^{2-} is _____.

- A. AlO
- B. Al_3O_2
- C. Al_2O_3
- D. AlO_3

A5 Topic: Ionic and Covalent Compound I IC1-LO3

The correct formula for ammonium phosphate is _____.

- A. NH_4PO_4
- B. $(\text{NH}_4)_3\text{PO}_4$
- C. $\text{NH}_4(\text{PO}_4)_3$
- D. $(\text{NH}_4)_2\text{PO}_4$

A6 Topic: Ionic and Covalent Compound I IC1-LO4

The Stock name of CuSO_4 is _____.

- A. copper sulfate
- B. copper(I) sulfate
- C. copper(II) sulfate
- D. cuprous sulfate

A7 Topic: Ionic and Covalent Compound I IC1-LO3

The formula of the nitrite ion is _____.

- A. NO_3^-
- B. NO_2^-
- C. N^{3-}
- D. NO_2^{2-}

A8 Topic: Ionic and Covalent Compound I IC1-LO5

The formula mass of CaCl_2 ($\text{Ca} = 40.08$, $\text{Cl} = 35.45$) is _____.

- A. 75.53 amu
- B. 110.98 amu
- C. 146.43 amu
- D. 115.53 amu

A9 Topic: Ionic and Covalent Compound I IC1-LO3

Which compound is **not** ionic?

- A. KBr
- B. MgO
- C. NH_4NO_3
- D. SO_2

A10 Topic: Ionic and Covalent Compound I IC1-LO6

Ionic compounds conduct electricity when they are _____.

- A. solid
- B. dissolved in water or melted
- C. cooled
- D. ground into a powder

Part B · Structured questions

Show all working. The marks and Bloom's level are shown for each question.

Question B1 Topic: Ionic and Covalent Compound I IC1-L02 5 marks, C2

Magnesium chloride (MgCl_2) is used in some electrolyte replacement solutions.

(a) Write the electron configurations of a magnesium atom ($Z = 12$) and a chlorine atom ($Z = 17$). (2 marks) (b) Describe how the ions form by electron transfer. (2 marks) (c) Explain why the formula is MgCl_2 and not MgCl . (1 mark)

Question B2 Topic: Ionic and Covalent Compound I IC1-L04 5 marks, C2

A community pharmacy stocks several ionic compounds used as antacids and supplements. (a) Write the formulas of aluminium hydroxide, magnesium hydroxide and calcium carbonate. (3 marks) (b) Name NaHCO_3 and FeSO_4 . (2 marks)

Question B3 Topic: Ionic and Covalent Compound I IC1-L05 6 marks, C2

Epsom salt, magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), is used as a laxative. (a) Calculate its formula mass. (5 marks) (b) Calculate the percentage of its mass that is water. (1 mark) ($\text{Mg} = 24.31$, $\text{S} = 32.07$, $\text{O} = 16.00$, $\text{H} = 1.01$)

Question B4 Topic: Ionic and Covalent Compound I IC1-L04 6 marks, C2

Iron forms two chlorides, FeCl_2 and FeCl_3 . For each compound, (a) determine the charge on the iron ion, (b) give the Stock name, and (c) give the common name.

Question B5 Topic: Ionic and Covalent Compound I IC1-L06 5 marks, C3

Normal saline (0.9% NaCl) is given by IV infusion. (a) Explain why solid sodium chloride does not conduct electricity but normal saline does. (2 marks) (b) Write the dissociation equation for NaCl dissolving in water, with state symbols. (2 marks) (c) Explain why sodium chloride melts at about 800°C while water melts at 0°C . (1 mark)

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Ionic and Covalent Compound I IC1-LO1

Atoms are especially stable when their valence shell contains _____ electrons.

- A. 2
- B. 6
- C. 8**
- D. 18

This is the octet rule (small atoms such as H follow the duet rule).

A2 Topic: Ionic and Covalent Compound I IC1-LO2

A calcium atom (Group 2A) forms the ion _____.

- A. Ca^+
- B. Ca^{2+}**
- C. Ca^{2-}
- D. Ca^-

Calcium loses its two valence electrons.

A3 Topic: Ionic and Covalent Compound I IC1-L02

Which pair of species is isoelectronic?

A. Na^+ and Ne

B. Na^+ and Ar

C. Cl^- and Ne

D. K^+ and Ne

Na^+ has 10 electrons, the same as neon.

A4 Topic: Ionic and Covalent Compound I IC1-L03

The correct formula for the compound of Al^{3+} and O^{2-} is _____.

A. AlO

B. Al_3O_2

C. Al_2O_3

D. AlO_3

Two Al^{3+} (6+) balance three O^{2-} (6-).

A5 Topic: Ionic and Covalent Compound I IC1-L03

The correct formula for ammonium phosphate is _____.

A. NH_4PO_4

B. $(\text{NH}_4)_3\text{PO}_4$

C. $\text{NH}_4(\text{PO}_4)_3$

D. $(\text{NH}_4)_2\text{PO}_4$

Three NH_4^+ balance one PO_4^{3-} .

A6 Topic: Ionic and Covalent Compound I IC1-L04

The Stock name of CuSO_4 is _____.

A. copper sulfate

B. copper(I) sulfate

C. copper(II) sulfate

D. cuprous sulfate

SO_4^{2-} is 2-, so copper is 2+: copper(II), or cupric.

A7 Topic: Ionic and Covalent Compound I IC1-LO3

The formula of the nitrite ion is _____.

- A. NO_3^-
- B. NO_2^-**
- C. N^{3-}
- D. NO_2^{2-}

-ite has one fewer oxygen than -ate (nitrate is NO_3^-).

A8 Topic: Ionic and Covalent Compound I IC1-LO5

The formula mass of CaCl_2 (Ca = 40.08, Cl = 35.45) is _____.

- A. 75.53 amu
- B. 110.98 amu**
- C. 146.43 amu
- D. 115.53 amu

$40.08 + 2 \times 35.45 = 110.98$ amu.

A9 Topic: Ionic and Covalent Compound I IC1-LO3

Which compound is **not** ionic?

- A. KBr
- B. MgO
- C. NH_4NO_3
- D. SO_2**

SO_2 contains only nonmetals and no polyatomic ion.

A10 Topic: Ionic and Covalent Compound I IC1-LO6

Ionic compounds conduct electricity when they are _____.

- A. solid
- B. dissolved in water or melted**
- C. cooled
- D. ground into a powder

The ions must be free to move.

Part B · Structured questions

Question B1 Topic: Ionic and Covalent Compound I IC1-LO2 5 marks, C2

Magnesium chloride (MgCl_2) is used in some electrolyte replacement solutions.

(a) Write the electron configurations of a magnesium atom ($Z = 12$) and a chlorine atom ($Z = 17$). (2 marks) (b) Describe how the ions form by electron transfer. (2 marks) (c) Explain why the formula is MgCl_2 and not MgCl . (1 mark)

(a) Mg: $1s^2 2s^2 2p^6 3s^2$. Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$. (b) Mg loses its two 3s electrons to become Mg^{2+} (octet in the second shell). Each Cl gains one electron to become Cl^- (octet in the third shell), so the two electrons go to two chlorine atoms. (c) One Mg^{2+} ($2+$) needs two Cl^- ($2 \times 1-$) for charge balance, so the formula is MgCl_2 .

Step	Marking criteria	Marks
1	Mg: $1s^2 2s^2 2p^6 3s^2$	1
2	Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$	1
3	Mg loses 2 electrons $\rightarrow \text{Mg}^{2+}$	1
4	Each Cl gains 1 electron $\rightarrow \text{Cl}^-$ (two Cl atoms needed)	1
5	Charge balance: one $2+$ balanced by two $1-$ $\rightarrow \text{MgCl}_2$	1
	Total	5

Question B2 Topic: Ionic and Covalent Compound I IC1-L04 5 marks, C2

A community pharmacy stocks several ionic compounds used as antacids and supplements. (a) Write the formulas of aluminium hydroxide, magnesium hydroxide and calcium carbonate. (3 marks) (b) Name NaHCO_3 and FeSO_4 . (2 marks)

(a) $\text{Al}^{3+} + 3 \text{OH}^- \rightarrow \text{Al}(\text{OH})_3$; $\text{Mg}^{2+} + 2 \text{OH}^- \rightarrow \text{Mg}(\text{OH})_2$; $\text{Ca}^{2+} + \text{CO}_3^{2-} \rightarrow \text{CaCO}_3$.

(b) NaHCO_3 = sodium bicarbonate (sodium hydrogen carbonate). FeSO_4 : SO_4^{2-} is $2-$, so iron is $2+$ → iron(II) sulfate (ferrous sulfate).

Step	Marking criteria	Marks
1	$\text{Al}(\text{OH})_3$ (brackets and subscript 3)	1
2	$\text{Mg}(\text{OH})_2$	1
3	CaCO_3	1
4	NaHCO_3 = sodium bicarbonate / sodium hydrogen carbonate	1
5	FeSO_4 = iron(II) sulfate / ferrous sulfate	1
	Total	5

Question B3

Topic: Ionic and Covalent Compound I

IC1-L05

6 marks, C2

Epsom salt, magnesium sulfate heptahydrate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), is used as a laxative. (a) Calculate its formula mass. (5 marks) (b) Calculate the percentage of its mass that is water. (1 mark) ($\text{Mg} = 24.31$, $\text{S} = 32.07$, $\text{O} = 16.00$, $\text{H} = 1.01$)

$\text{MgSO}_4 = 24.31 + 32.07 + 4 \times 16.00 = 120.38 \text{ amu}$. $\text{H}_2\text{O} = 2 \times 1.01 + 16.00 = 18.02 \text{ amu}$, so $7\text{H}_2\text{O} = 126.14 \text{ amu}$. Formula mass = $120.38 + 126.14 = 246.52 \text{ amu}$. % water = $126.14 \div 246.52 \times 100 = 51.17\%$.

Step	Marking criteria	Marks
1	Set-up for MgSO_4 : $24.31 + 32.07 + 4(16.00)$	1
2	$\text{MgSO}_4 = 120.38 \text{ amu}$	1
3	$\text{H}_2\text{O} = 18.02 \text{ amu}$	1
4	$7\text{H}_2\text{O} = 126.14 \text{ amu}$	1
5	Formula mass = 246.52 amu	1
6	% water = $126.14 / 246.52 \times 100 = 51.17\%$	1
	Total	6

Question B4 Topic: Ionic and Covalent Compound I IC1-LO4 6 marks, C2

Iron forms two chlorides, FeCl_2 and FeCl_3 . For each compound, (a) determine the charge on the iron ion, (b) give the Stock name, and (c) give the common name.

FeCl_2 : two $\text{Cl}^- = 2-$, so Fe is $2+$ $\rightarrow$ iron(II) chloride, ferrous chloride. FeCl_3 : three $\text{Cl}^- = 3-$, so Fe is $3+$ $\rightarrow$ iron(III) chloride, ferric chloride.

Step	Marking criteria	Marks
1	FeCl_2 : iron is $2+$ (two Cl^-)	1
2	FeCl_3 : iron is $3+$ (three Cl^-)	1
3	FeCl_2 = iron(II) chloride	1
4	FeCl_3 = iron(III) chloride	1
5	FeCl_2 = ferrous chloride	1
6	FeCl_3 = ferric chloride	1
	Total	6

Question B5 Topic: Ionic and Covalent Compound I IC1-LO6 5 marks, C3

Normal saline (0.9% NaCl) is given by IV infusion. (a) Explain why solid sodium chloride does not conduct electricity but normal saline does. (2 marks) (b) Write the dissociation equation for NaCl dissolving in water, with state symbols. (2 marks) (c) Explain why sodium chloride melts at about 800 °C while water melts at 0 °C. (1 mark)

(a) In the solid, the ions are locked in place in the crystal lattice and cannot move. In solution the lattice has broken up, the ions move freely and carry the current.

(b) $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$. (c) Each ion is attracted to many oppositely charged neighbours, so melting NaCl means breaking many strong ionic attractions; water is a molecular compound, and much less energy is needed.

Step	Marking criteria	Marks
1	Solid: ions fixed in the lattice, cannot move	1
2	Solution: ions free to move and carry charge	1
3	Correct species: Na^+ and Cl^- from NaCl	1
4	Correct state symbols (s) → (aq)	1
5	Strong ionic attractions in all directions need much energy to break	1
	Total	5

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